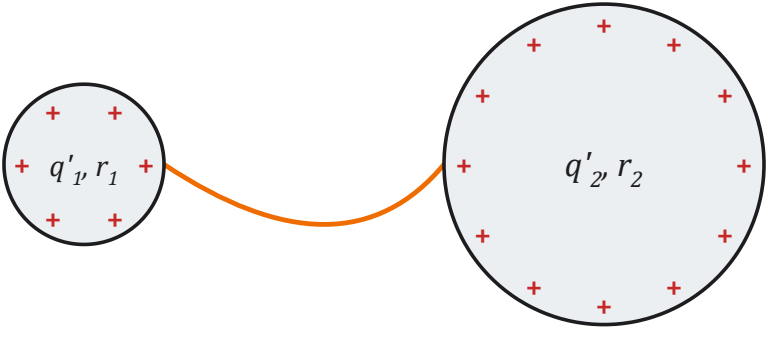

$$V'_1 = V'_2 \Rightarrow q'_1 / r_1 = q'_2 / r_2$$